Amazon EventBridge EC2 State Change Notification Lab

To Begin with the Lab

Summary of the Lab

In this lab, **Amazon EventBridge** was used to monitor **Amazon EC2** state-change events and send notifications to an **Amazon SNS** topic. Two EC2 instances were available as the event source, and an SNS topic with an email subscriber was created as the target. First, a broad EventBridge rule was created to capture all EC2 events, which resulted in notifications for multiple instance state transitions such as stopping, stopped, pending, and running. Then the rule was refined to filter only one specific EC2 instance and only the **Stopped** state. This showed how EventBridge rules can reduce noise and send notifications only for the events that matter. The lab ended by confirming that the email notification was triggered only when the selected instance reached the configured state.

Understanding EventBridge

Amazon EventBridge is an event bus service that receives events from AWS services and routes them to targets based on rules. It exists so you can react to AWS events without constantly polling resources or writing custom scripts. In this lab, EC2 state-change events were sent to the default event bus, and EventBridge rules decided whether the event should be forwarded to SNS. The key idea is that EventBridge does not do anything by itself unless a rule matches the incoming event pattern.

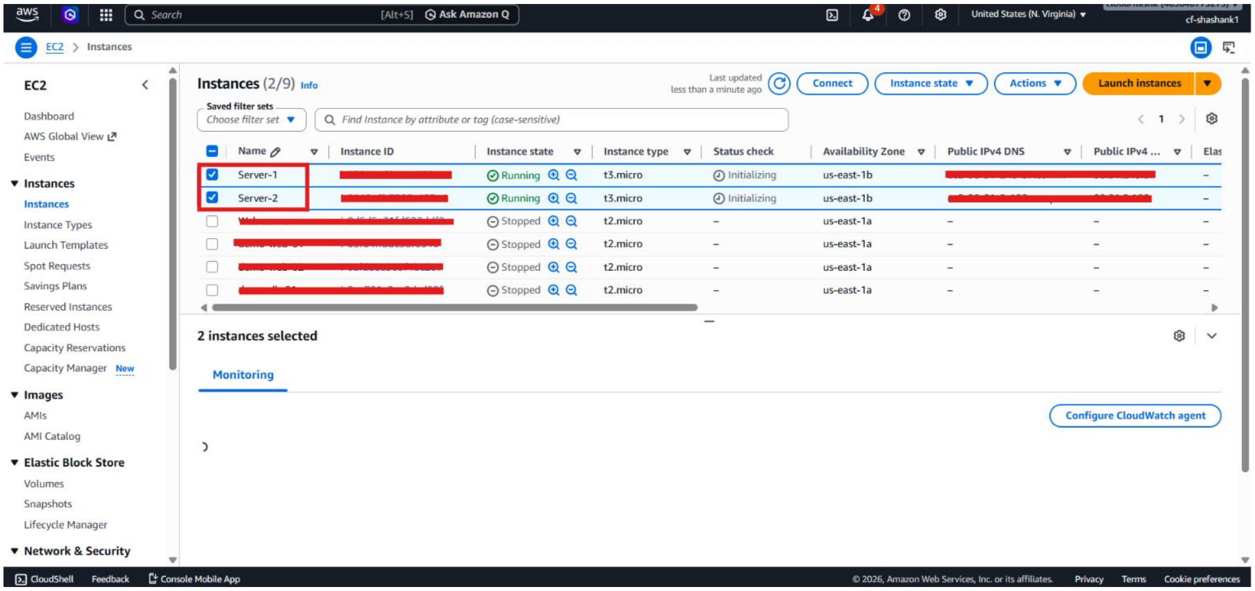
Example

If an EC2 instance changes from **running** to **stopping**, EventBridge can detect that state-change event and send an SNS notification. Later, if you only want an alert when a specific instance reaches **stopped**, you can filter the rule so that only that instance and that state trigger the message. That way, you avoid receiving emails for every minor EC2 state transition.

Lab Steps

Step 1 – Prepare the EC2 Event Source

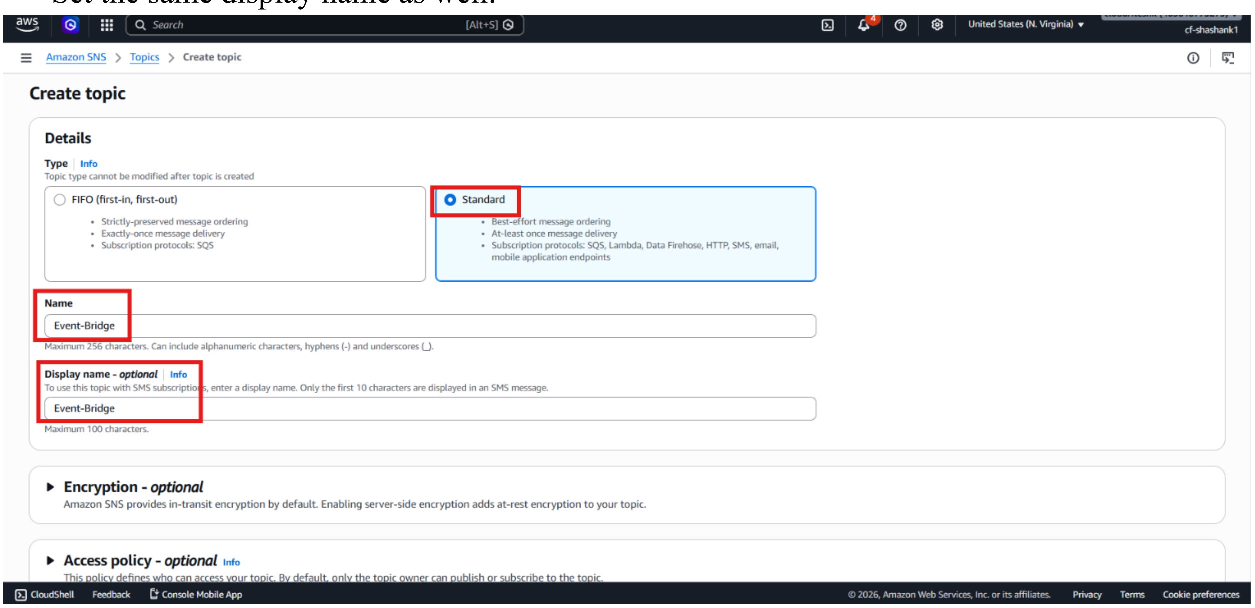
- Sign in to the AWS Management Console.
- Open **Amazon EC2**.
- Launch two small EC2 instances.



- Keep them in the default VPC.
- Use **t3.micro** or another free-tier eligible instance type.
- Keep both instances running so they can generate events.

Step 2 – Create the SNS Topic

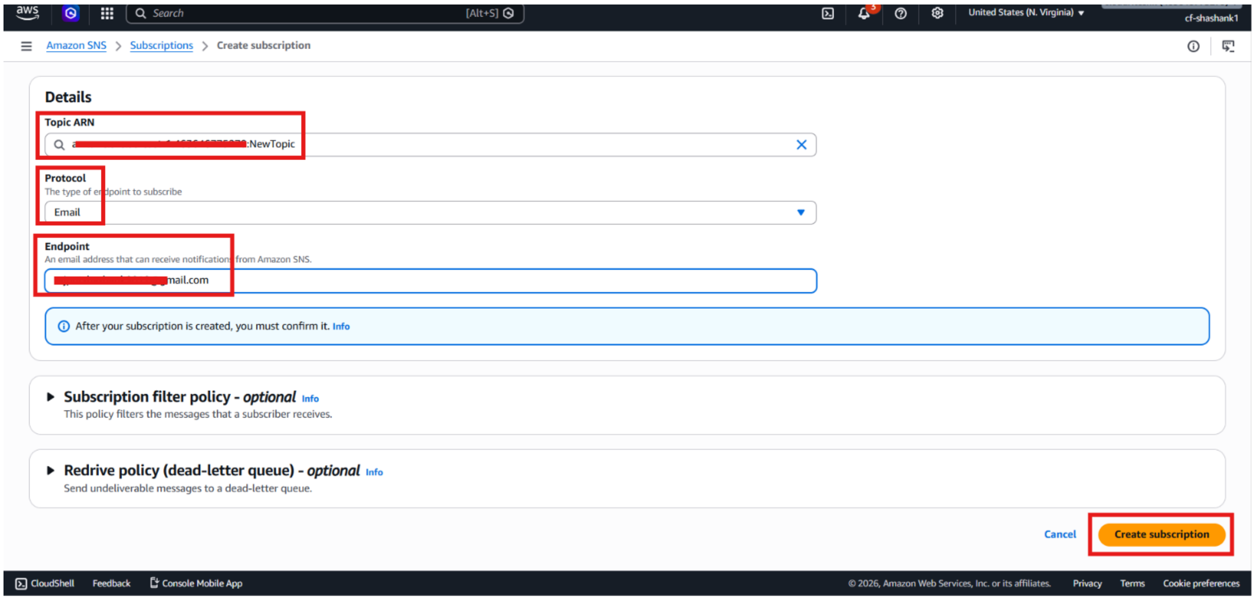
- Open **Amazon SNS**.
- Go to **Topics**.
- Click **Create topic**.
- Select **Standard**.
- Enter the topic name: Event-Bridge
- Set the same display name as well.



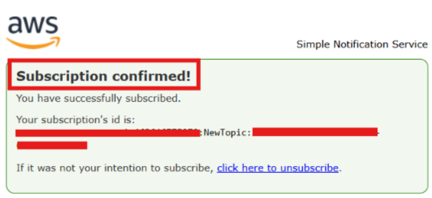
- Create the topic.

Step 3 – Add an Email Subscriber

- Open the SNS topic.
- Click **Create subscription**.
- Select **Email**.
- Enter your email address as the endpoint.
- Create the subscription.



- Open the confirmation email.
- Click **Confirm subscription**.

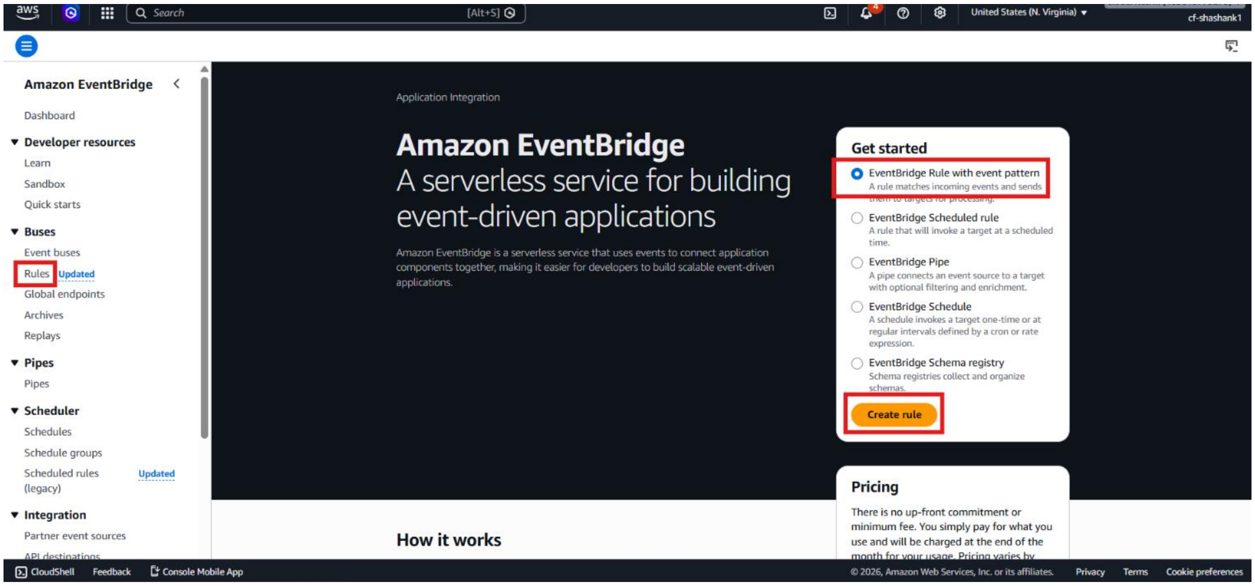


- Verify that the subscriber status becomes **Confirmed**.

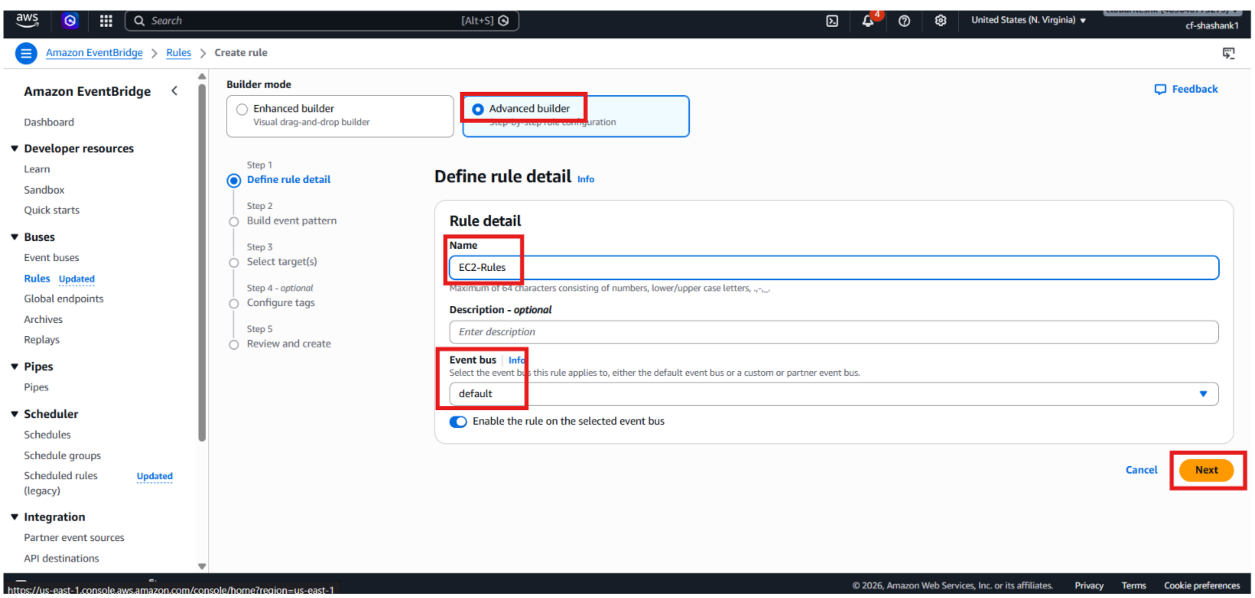
Step 4 – Create an EventBridge Rule for All EC2 Events

- Open **Amazon EventBridge**.

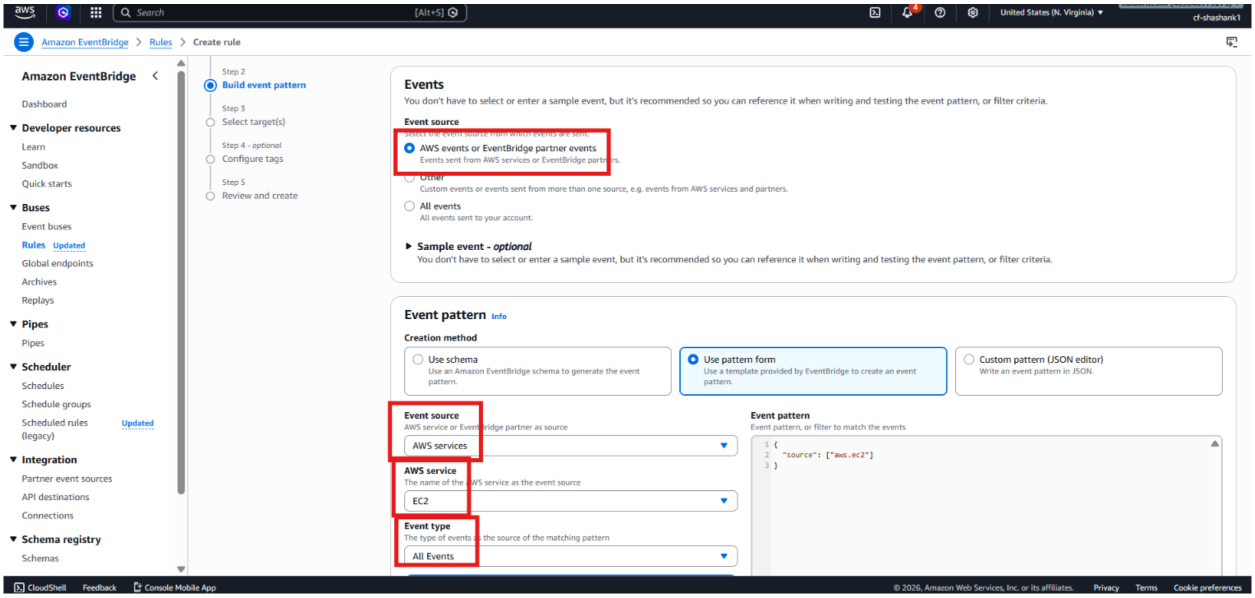
- Go to **Rules**.
- Click **Create rule**.



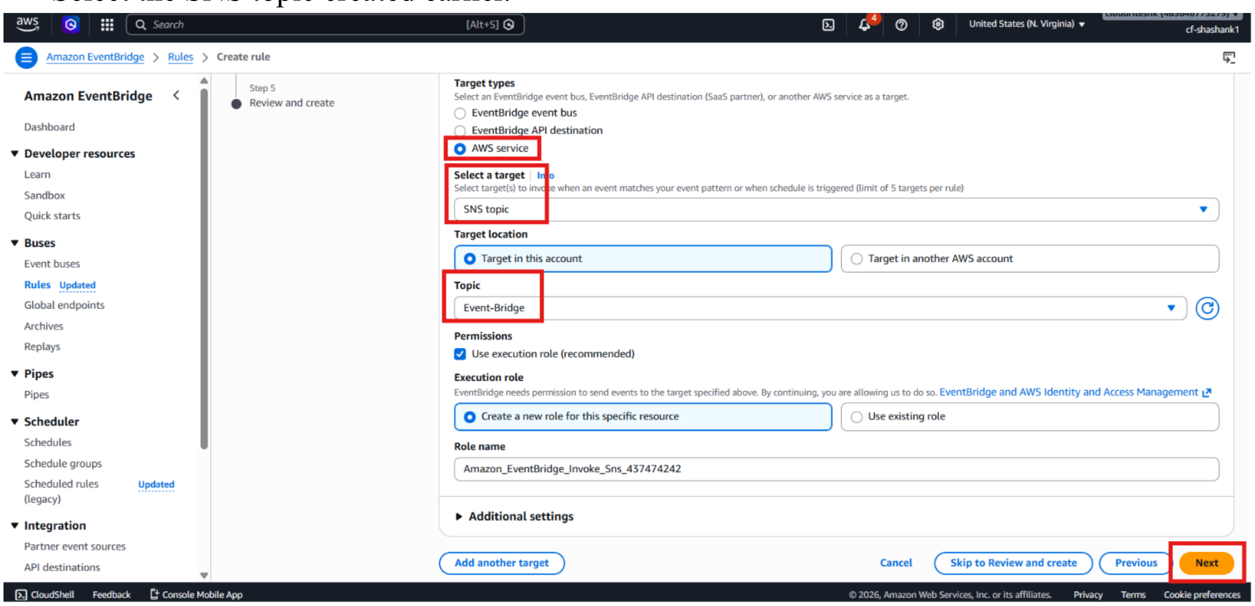
- Choose **Advanced builder**
- Enter the rule name: EC2-Rules
- Use the **Default event bus**.



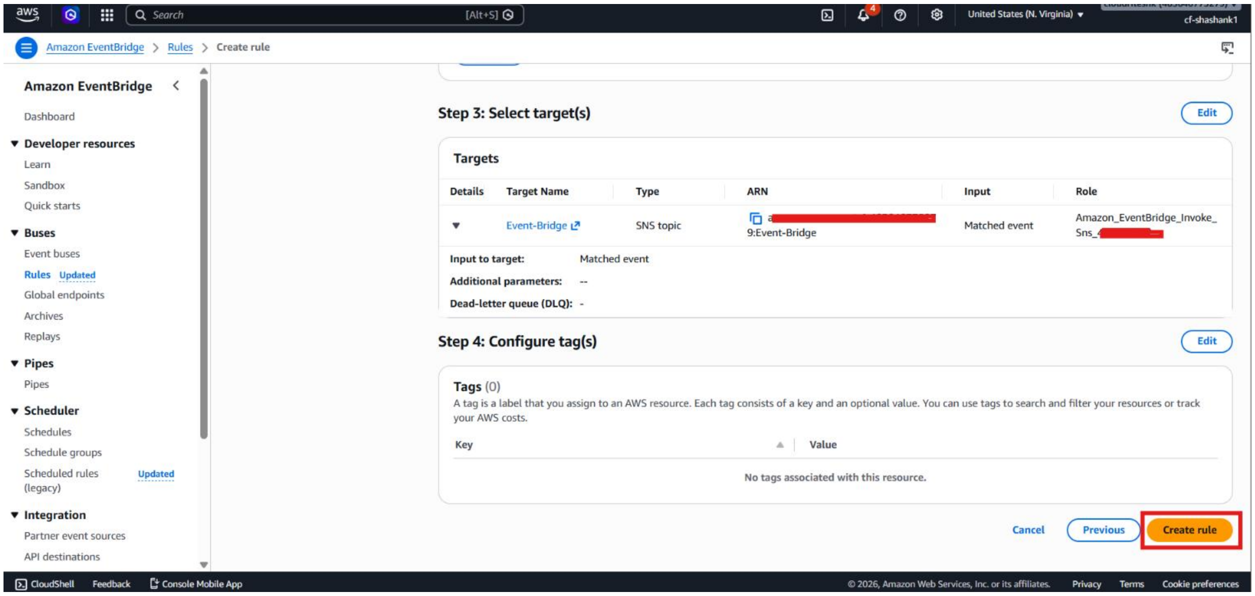
- Choose the AWS event pattern option.
- Select **AWS services**.
- Choose **EC2** as the event source.
- Select **All events**.



- Continue to the target section.
- Choose **SNS topic** as the target.
- Select the SNS topic created earlier.

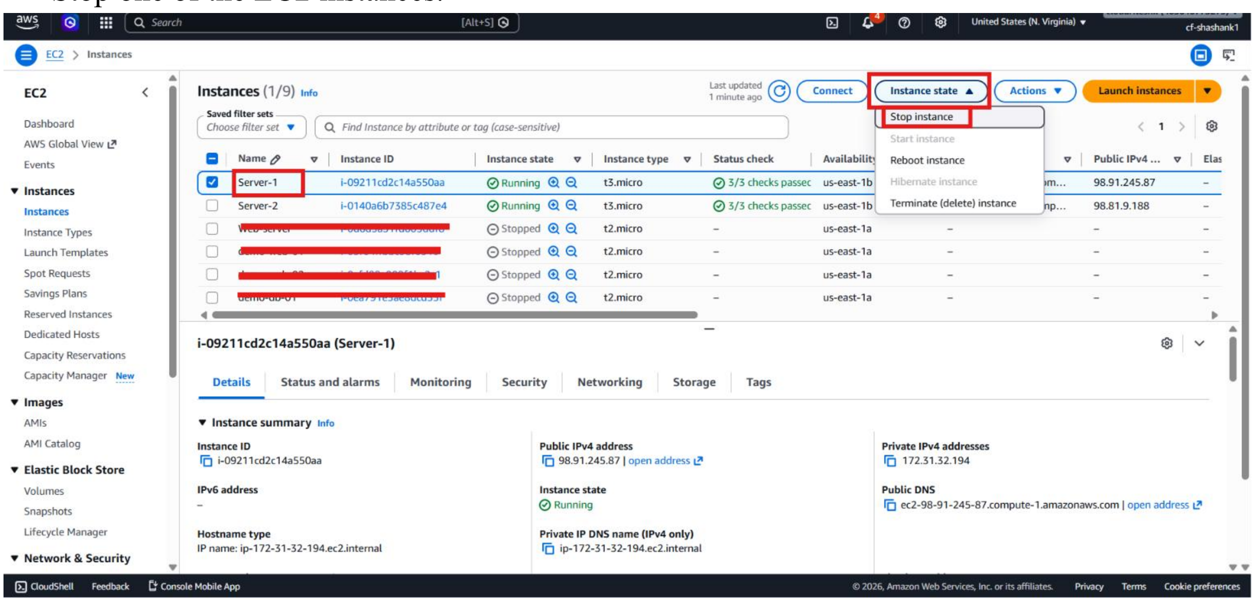


- Allow EventBridge to create the required IAM role automatically.
- Create the rule.

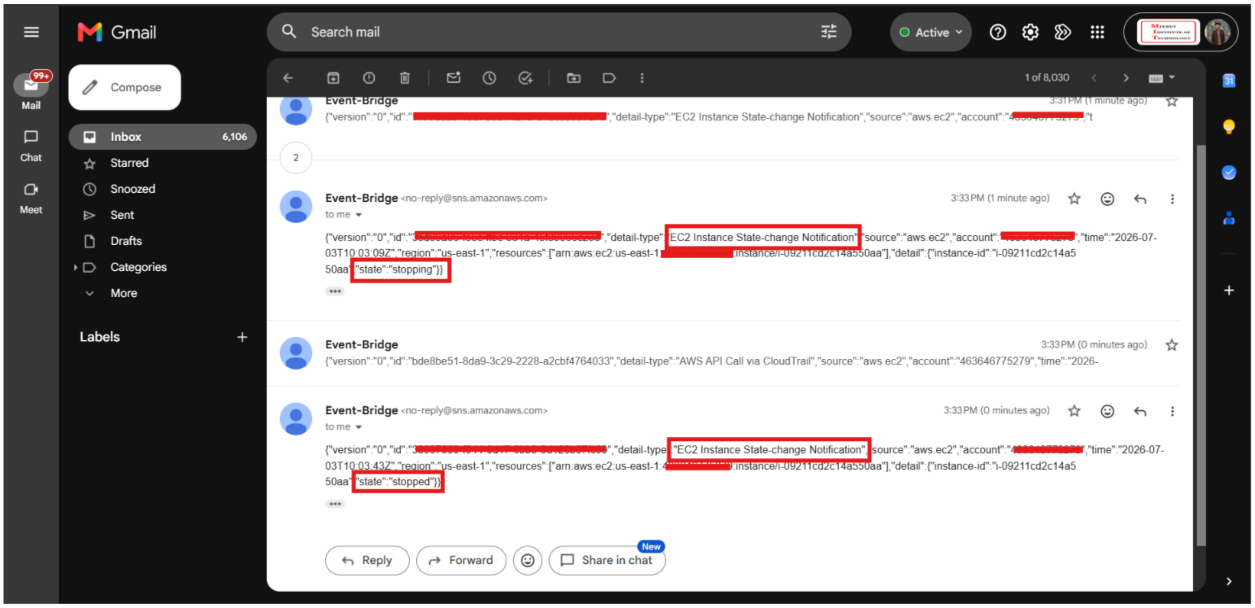


Step 5 – Test the Broad Rule

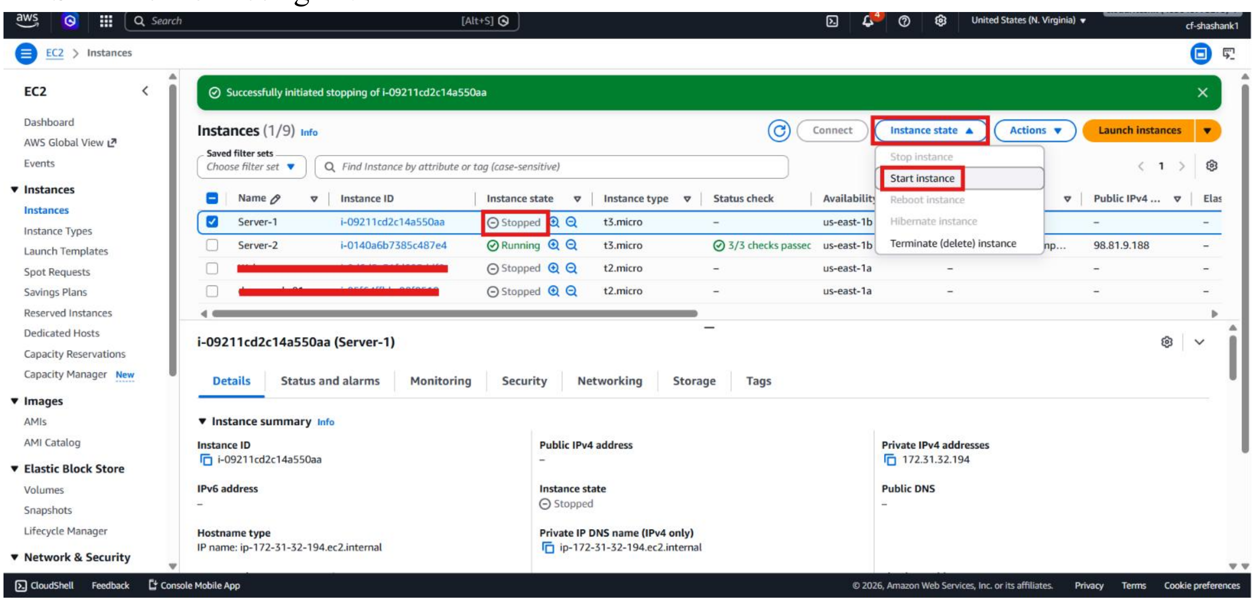
- Stop one of the EC2 instances.



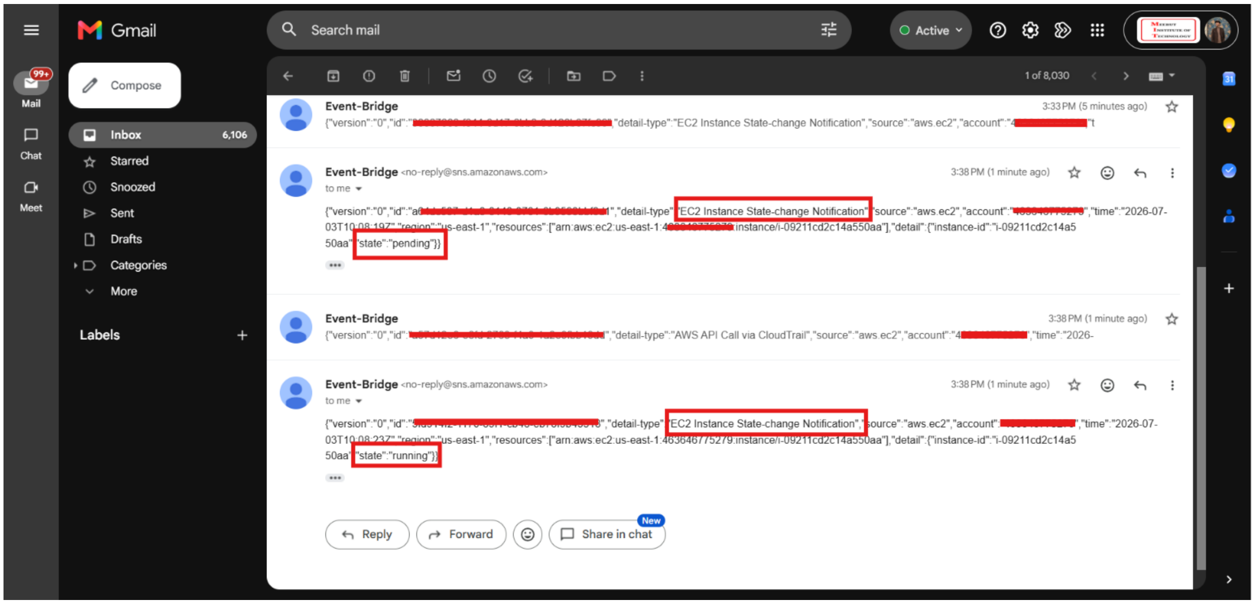
- Observe that EventBridge sends notifications for multiple state changes.



- Start the instance again.



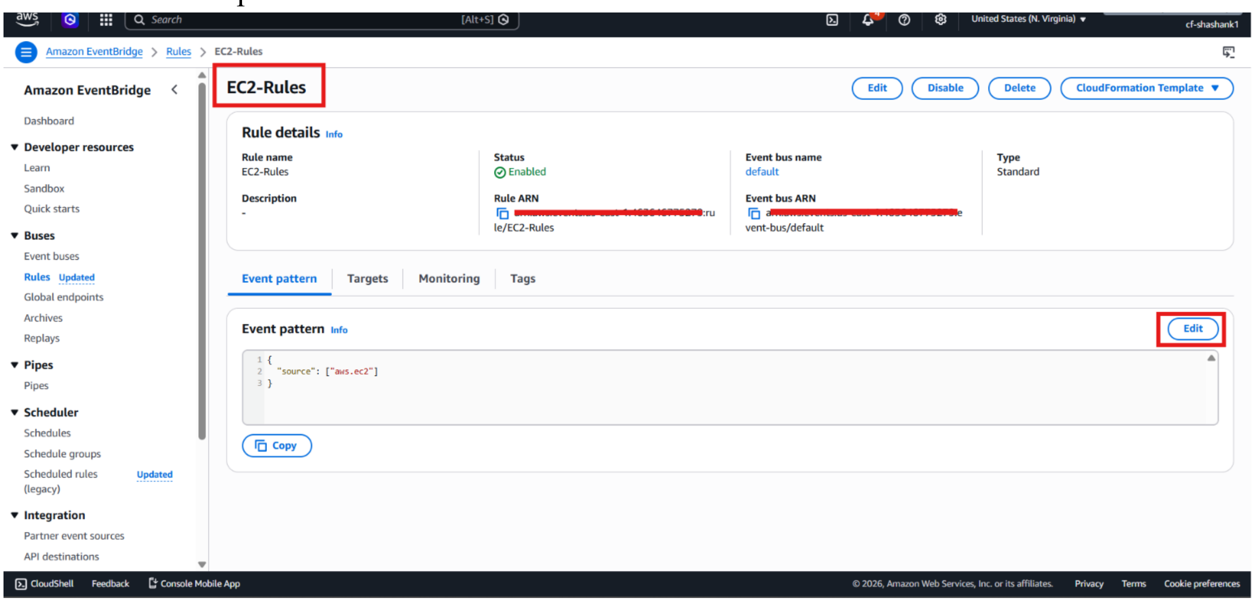
- Observe that notifications are also sent for the start sequence.



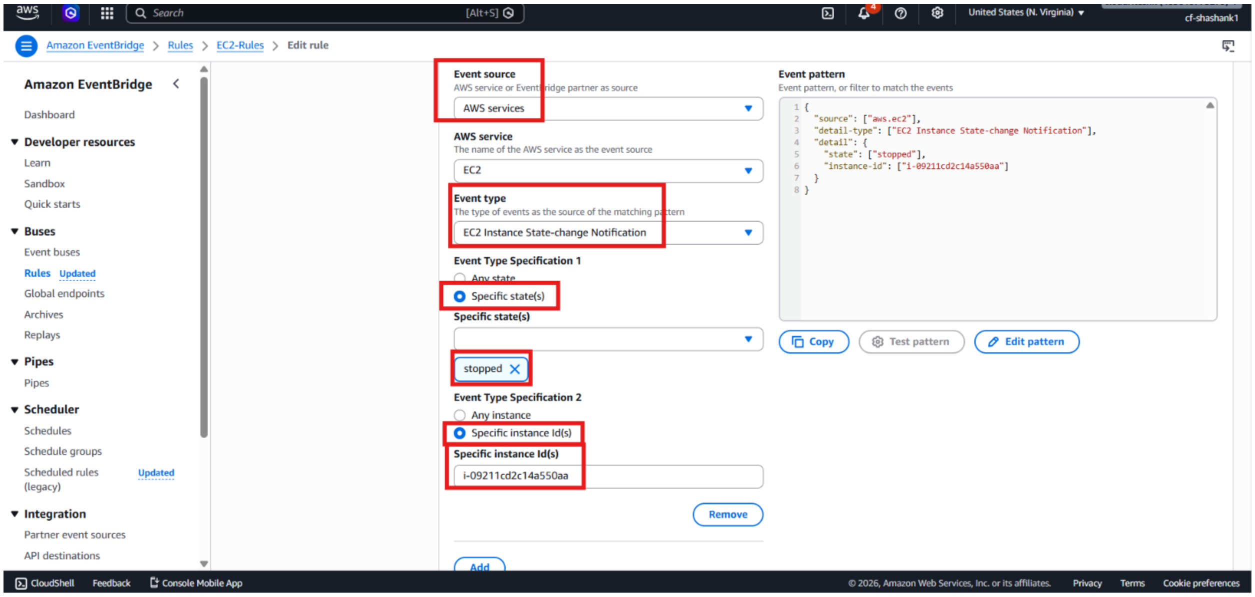
- Confirm that the rule is matching every EC2 event.

Step 6 – Refine the Rule to One Instance and One State

- Open the existing EventBridge rule.
- Edit the event pattern.



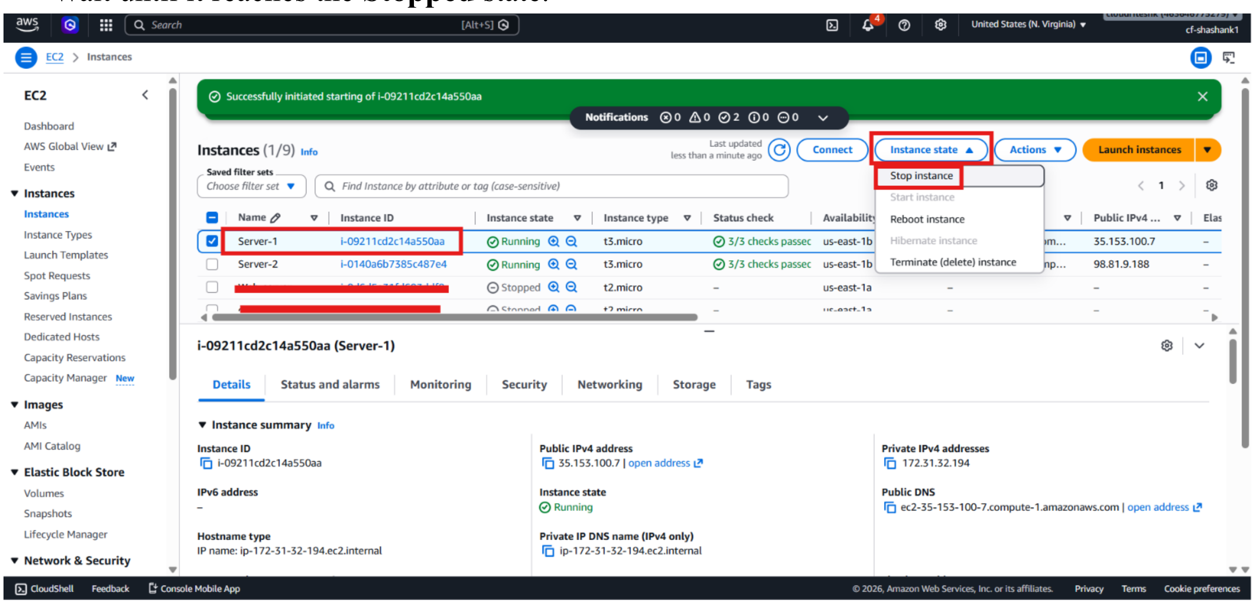
- Change the event type to **EC2 instance state-change notification**.
- Set the desired state to **Stopped**.
- Add the specific EC2 instance ID you want to monitor.



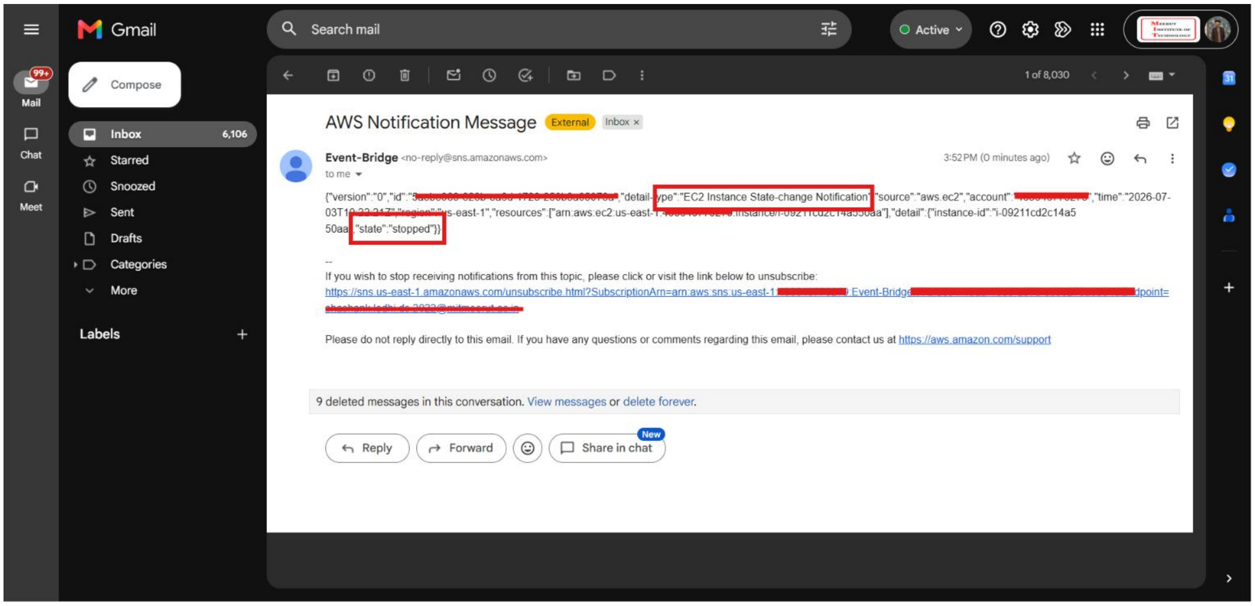
- Save the updated rule.

Step 7 – Test the Filtered Rule

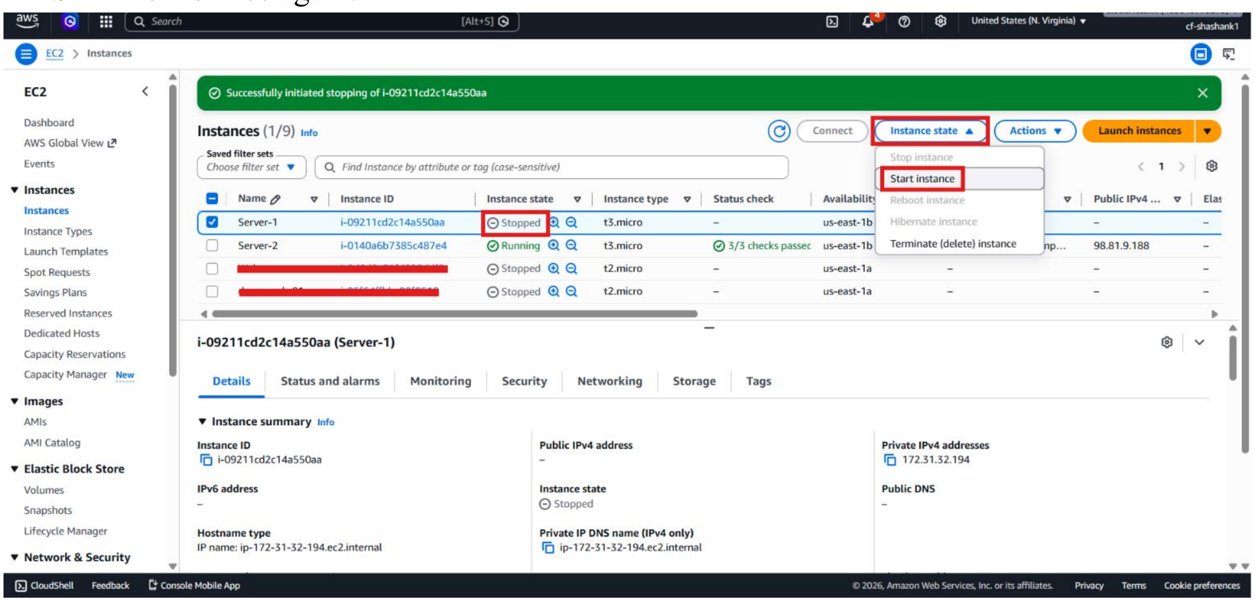
- Stop the selected EC2 instance.
- Wait until it reaches the **Stopped** state.



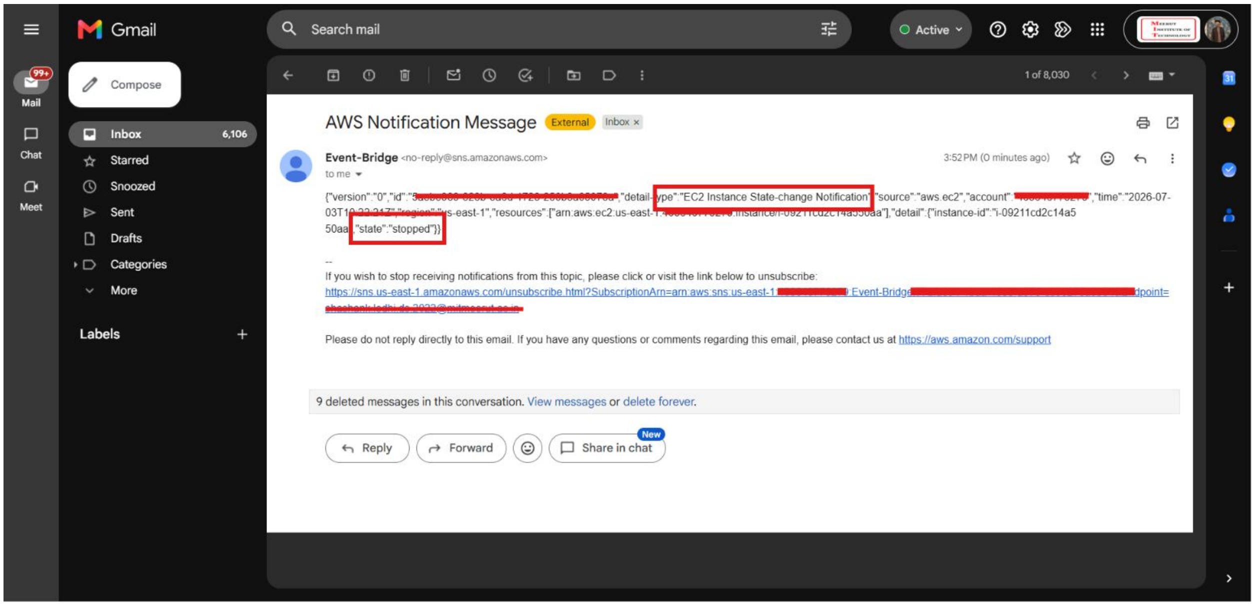
- Check your email inbox.
- Confirm that only one notification is received.



- Start the instance again.



- Confirm that no extra notification is received because the rule only matches the **Stopped** state for that specific instance.



- Stop the second EC2 instance.
- Confirm that no notification is sent because it is not part of the rule.

Verification

- Two EC2 instances were created successfully.
- An SNS topic and email subscriber were configured successfully.
- EventBridge was connected to the default event bus.
- The first rule captured all EC2 state changes.
- The rule was later refined to a single EC2 instance and the **Stopped** state.
- Email notifications were triggered only when the rule matched.
- The lab confirmed that EventBridge filtering works correctly.